

Probability Density Function and the CDF of a Continuous Variable

Contents

1. Overview	2
2. The probability density function	3
3. Why the y axis cannot be probability	3
4. Area under the curve is probability	4
5. What the density at a point means	5
6. Famous PDFs	6
7. The CDF of a continuous variable	6
8. How the PDF and CDF are linked	8
9. Summary	9
Sources	9
10. Key terms	9

Where this fits: Pipeline map steps 0 (Foundations), 3 (Understand data). See the [Course map](#).



- **Builds on:** Probability distributions ([Note 240](#)); Probability mass function (PMF) ([Note 241](#)).
- **Leads to:** Density estimation ([Note 243](#)); Normal distribution ([Note 250](#)); Standard normal and the z-table ([Note 251](#)); P-values ([Note 300](#)).
- **Compare with:** Frequency tables ([Note 223](#)); Bernoulli and binomial distributions ([Note 241](#)); Pareto distribution and power laws ([Note 262](#)).

1. Overview

Key point: For a continuous random variable the probability of any single exact value is 0, so the PDF's height is a density, and probabilities are areas under it; the CDF is the area to the left of x .

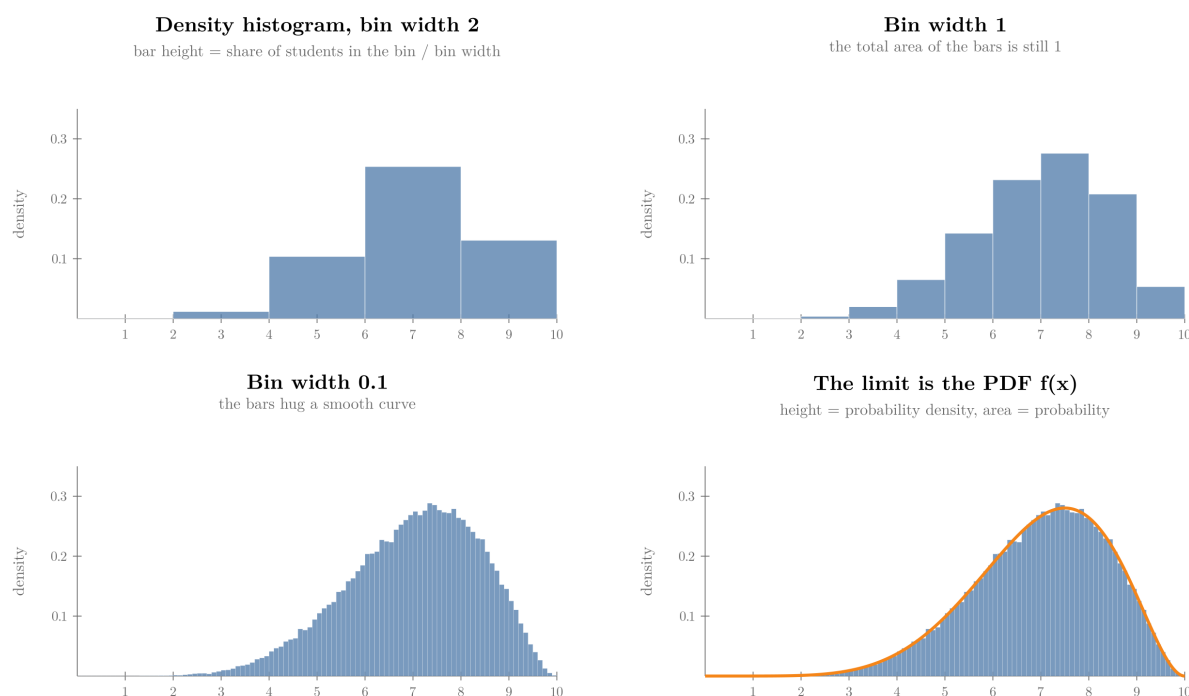


Figure 1: Density histograms with narrower and narrower bins settle onto the PDF curve

Figure 1 shows where the PDF comes from. We take 100,000 CGPAs and draw a density histogram, in which each bar's **area** is the share of students in its bin. As the bins get narrower, the bars hug a smooth curve: the probability density function.

We met this curve in the [univariate analysis Note](#) (section 7) as the KDE drawn over a histogram, where we noted that its height is a density, not a probability. This Note explains why, shows how areas give probabilities, and builds the CDF of a continuous variable. How the curve is estimated from data is the topic of the [density estimation Note](#).

2. The probability density function

Key point: A PDF is the probability distribution function of a continuous random variable; unlike a PMF, its y axis is probability density, not probability.

The probability density function (PDF) is a mathematical function that describes the probability distribution of a **continuous** random variable (see the [random variables and distributions Note](#)). Its graph is an unbroken curve (Figure 2, left).

The x axis works as before: it holds the values of the variable, for example CGPAs from 0 to 10, now with every decimal in between. The difference is the y axis. A PDF differs from a PMF in two ways:

	PMF	PDF
Random variable	discrete	continuous
y axis	probability	probability density
Probability of a range	sum of the bar heights	area under the curve

The density plot of the [univariate analysis Note](#) (section 7) already met this: its curve's height is a density, and probability is the area under it.

The second row is the one to remember. On a PMF we read probabilities straight off the graph. On a PDF the height is a **probability density**, and the rest of this Note is about what that means.

3. Why the y axis cannot be probability

Key point: A continuous variable has infinitely many possible values, so the probability of any one exact value is 0; probabilities only exist for ranges.

Pick one student and ask the probability that their CGPA is exactly 7.912. Between 0 and 10 there are infinitely many possible values: 7.912, 7.9121, 7.91201 and so on. Spreading a total probability of 1 over infinitely many values leaves 0 for each one:

$$P(X = 7.912) = 0$$

The same shows up in real data. Among 100 students, perhaps one has a CGPA of 7.912 to three decimals. Ask for 7.912959 and almost certainly nobody has it; the finer we ask, the closer the share gets to 0.

Shrinking a range shows it with numbers. For the CGPA curve of Figure 2, the probability of a CGPA between 8 and $8 + h$ falls to 0 as the width h shrinks:

Width h	$P(8 \leq X \leq 8 + h)$	$P(8 \leq X \leq 8 + h) / h$
1	0.2088	0.2088
0.1	0.0261	0.2607
0.01	0.00264	0.2639
0.001	0.000264	0.2642

So a graph of “probability at each x ” would be flat at 0 everywhere and tell us nothing. The last column, however, settles on a number: 0.264. That number is the density at 8 (Section 5).

4. Area under the curve is probability

Key point: The total area under a PDF is 1; the area between two values a and b is the probability that the variable falls between them.

The whole area under the CGPA curve stands for the probability that a student’s CGPA is somewhere between 0 and 10. That is certain, so the **total area under every PDF is 1**.

A slice of the area gives a smaller probability. The area between 8 and 9 is the probability of a CGPA between 8 and 9 (Figure 2, left). Since the curve is not a rectangle, the area is found by **integration**, which adds up the area of infinitely many infinitely thin strips under the curve.

1. **In words:** the probability that X falls between a and b is the area under the PDF from a to b .
2. **Formula:**

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

The symbol $\int_a^b$ reads “the area from a to b under”, and dx marks x as the variable along the horizontal axis.

3. **Example:** for the CGPA curve,

$$P(8 \leq X \leq 9) = \int_8^9 f(x) dx = 0.209$$

About 21% of students have a CGPA between 8 and 9.

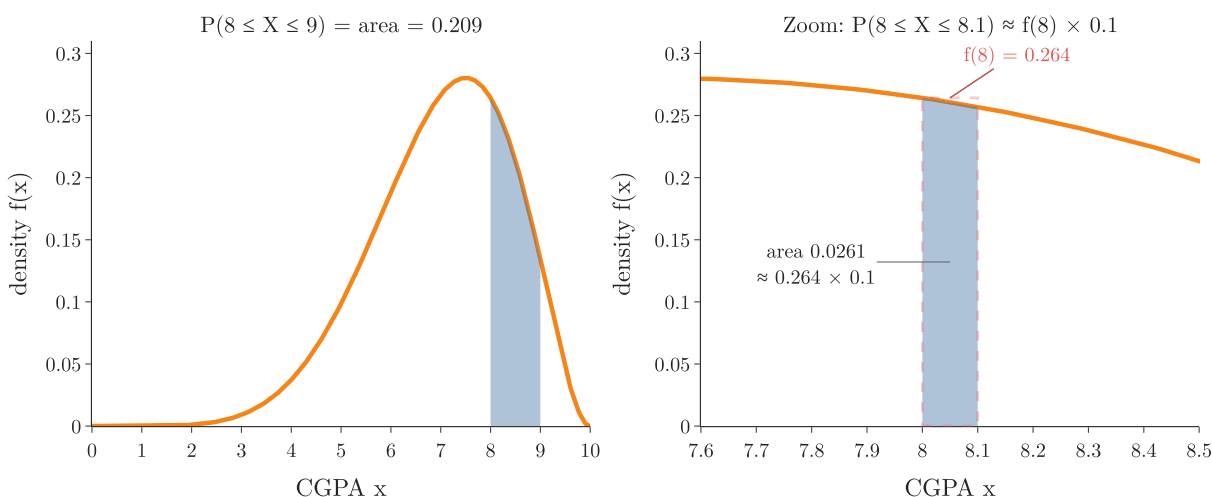


Figure 2: Probability is area under the PDF: between 8 and 9 (left), and a thin slice from 8 to 8.1 (right)

Because a single point has zero width, it has zero area. So for a continuous variable it makes no difference whether the ends are included: $P(8 \leq X \leq 9) = P(8 < X < 9)$.

Python: The area from a known PDF, two ways.

```
from scipy import stats, integrate
```

```
# CGPA curve used in this Note: 10 x Beta(7, 3)
```

```
cgpa = stats.beta(7, 3, scale=10)
```

```
# area by numerical integration
```

```
integrate.quad(cgpa.pdf, 8, 9)[0] # 0.2088
```

```
# the same area from the CDF (Section 7)
```

```
cgpa.cdf(9) - cgpa.cdf(8) # 0.2088
```

The curve is a scaled beta distribution, chosen because it lives exactly on 0 to 10; any smooth curve with area 1 would do.

5. What the density at a point means

Key point: The density at x is the probability per unit of x near x : over a narrow width h , the probability is about $f(x) \times h$.

The density $f(x)$ is what probability turns into when we divide by the width of a narrow range (the last column of the table in Section 3). Turned around, a thin strip under the curve is almost a rectangle of height $f(x)$ and width h (Figure 2, right).

1. **In words:** the probability of landing in a narrow range is about the density times the width of the range.
2. **Formula:**

$$P(x \leq X \leq x + h) \approx f(x) \times h \quad \text{for small } h$$

3. **Example:** the density at 8 is $f(8) = 0.264$. For the range 8 to 8.01:

$$P(8 \leq X \leq 8.01) \approx 0.264 \times 0.01 = 0.00264$$

The exact area is 0.002639; the smaller h , the better the match.

So for practical purposes a higher curve still means “more likely around here”, and we can compare densities at two points the way we compare probabilities. Strictly, though, the height is never a probability itself.

This is the same idea as the density histogram of Figure 1. A density bar has height “share of the data in the bin divided by the bin width”, so its area is the share. As the bins narrow, those heights become $f(x)$.

Extra: Because a density is probability **per unit**, it can be larger than 1 (see the [Gaussian Naive Bayes Note](#)). A uniform distribution on 0 to 0.5 has height 2 everywhere: width 0.5 times height 2 gives the required area of 1. A probability can never exceed 1; a density can.

6. Famous PDFs

Key point: The normal and log-normal distributions are famous continuous distributions with known PDF formulas; Poisson, often listed with them, is discrete.

The famous continuous distributions of the [random variables and distributions Note](#) (Figure 4) each have a PDF formula:

- **Normal distribution:** parameters μ (mean, location) and σ (standard deviation, scale). Much natural data follows it. Its PDF, worked through in the [Gaussian Naive Bayes Note](#), is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- **Log-normal distribution:** looks like a normal curve pushed to the left, with a long right tail. Its parameters are also μ and σ , those of the logarithm of the variable (SciPy `lognorm` docs).

The **Poisson distribution** (parameter λ) is often listed beside them, but it counts events (0, 1, 2, ...), so it is discrete and has a PMF, not a PDF. SciPy, for example, gives `poisson` a `pmf` and no `pdf`. Each of these distributions gets its own Note later.

7. The CDF of a continuous variable

Key point: The CDF $F(x) = P(X \leq x)$ is the area under the PDF to the left of x ; it rises smoothly from 0 to 1.

The CDF has the same definition as for a discrete variable (see the [PMF and discrete CDF Note](#)): the probability of a value at most x . For a continuous variable, “at most x ” is the area under the PDF from the far left up to x .

1. **In words:** the CDF at x is the whole area under the PDF to the left of x .
2. **Formula:**

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

The letter t runs along the axis up to x ; $-\infty$ means “from the far left”.

3. **Example:** adult heights in a group follow a normal distribution with mean 165 cm and standard deviation 10 cm. Half the area lies left of the mean, so $F(165) = 0.5$: half the people are 165 cm or shorter. The area left of 150 is $F(150) = 0.067$: about 7% are 150 cm or shorter.

Figure 3 puts the two functions one above the other. The same point means different things on each:

- **On the PDF at 165:** the height $f(165) = 0.0399$ is a density. Roughly, heights near 165 cm are the most common ones.
- **On the CDF at 165:** the height $F(165) = 0.5$ is a probability: the chance of 165 cm or less.

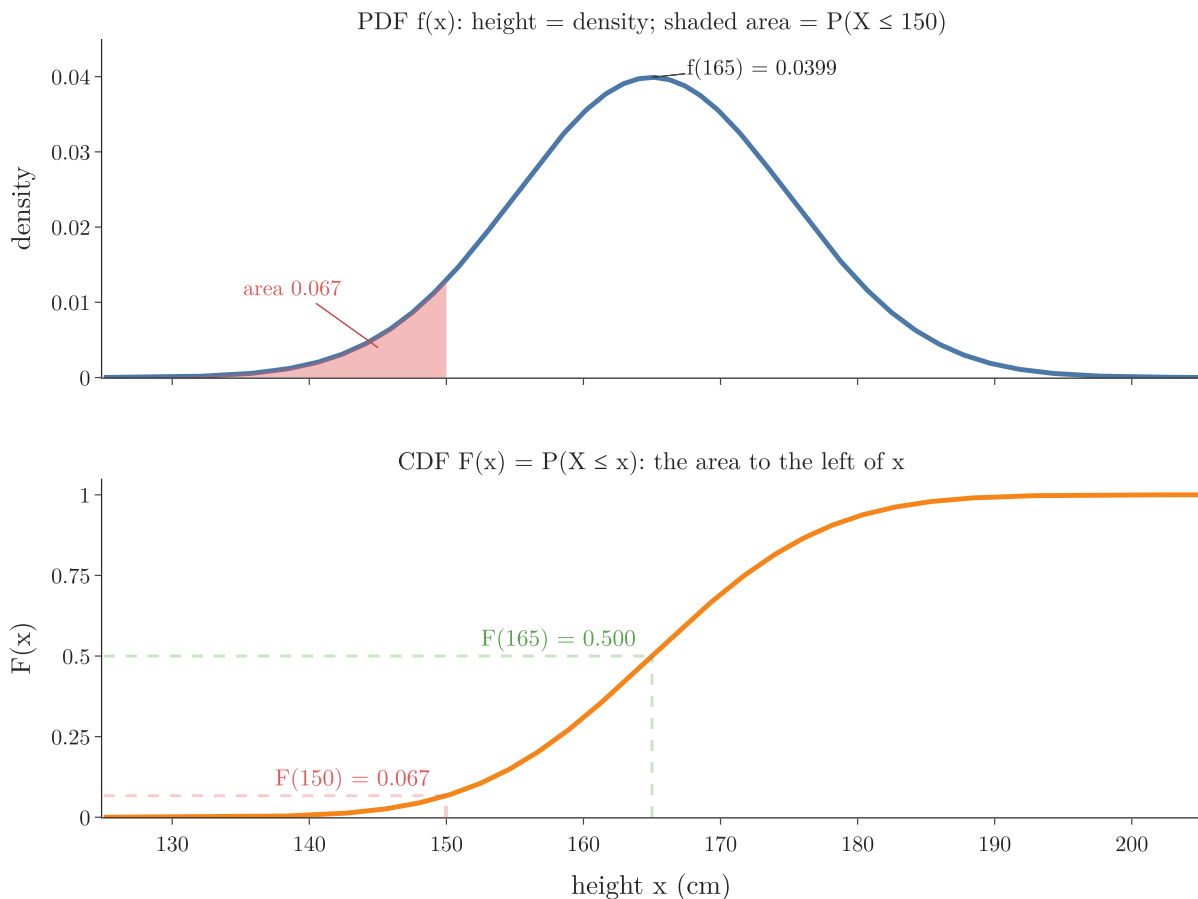


Figure 3: Heights: the PDF with the area left of 150 shaded (top) and the CDF that reads off such areas (bottom)

The CDF of a continuous variable has no steps: it rises smoothly, steepest where the PDF is highest.

Python: PDF and CDF values.

```
height = stats.norm(165, 10)
height.pdf(165)    # 0.0399, a density
height.cdf(165)    # 0.5, a probability
height.cdf(150)    # 0.067
```

Extra: The CDF turns any range into a subtraction: $P(a < X \leq b) = F(b) - F(a)$. Heights between 155 and 175 cm, one standard deviation either side of the mean:

$$P(155 < X \leq 175) = F(175) - F(155) = 0.841 - 0.159 = 0.683$$

This is the 68% of the 68-95-99.7 rule (see the [z-score outliers Note](#)).

8. How the PDF and CDF are linked

Key point: The area under the PDF gives the CDF, and the slope of the CDF gives the PDF; in calculus terms, integration goes one way and differentiation the other.

The two curves of Figure 3 carry the same information:

- **PDF to CDF:** the CDF at x is the area under the PDF up to x (Section 7). Collecting area is integration.
- **CDF to PDF:** where the CDF climbs steeply, much probability is packed into a short range, so the density is high. The slope of the CDF at x is the PDF at x . Taking a slope is differentiation.

1. **In words:** the steepness of the CDF at a point equals the height of the PDF there.

2. **Formula:**

$$f(x) = \frac{dF(x)}{dx} \approx \frac{F(x + 0.5) - F(x - 0.5)}{1}$$

The left side is the derivative, the exact slope. The right side measures the rise of the CDF over a step of 1 around x .

3. **Example:** for the heights at 165,

$$\frac{F(165.5) - F(164.5)}{1} = 0.0399 = f(165)$$

At 150 the same rise is 0.0130, and $f(150) = 0.0130$: the CDF is flatter there, and the PDF lower.

Calculus is not needed to use these ideas: libraries compute both functions. The link returns later: the z-table of the [standard normal Note](#) is a table of areas under the normal PDF, that is, of its CDF.

9. Summary

Idea	Formula	Example
Probability of one exact value	$P(X = x) = 0$	$P(X = 7.912) = 0$
Probability of a range	$\int_a^b f(x) dx$	$P(8 \leq X \leq 9) = 0.209$
Total area	$\int f(x) dx = 1$	whole CGPA curve
Density as probability per unit	$P(x \leq X \leq x + h) \approx f(x) h$	$0.264 \times 0.01 = 0.00264$
CDF	$F(x) = \int_{-\infty}^x f(t) dt$	$F(150) = 0.067$
PDF from CDF	$f(x) = dF/dx$	slope at 165 = 0.0399

- PMF: discrete, height is probability. PDF: continuous, height is density, area is probability.
- A density can exceed 1; a probability cannot.
- Poisson is discrete (PMF); normal and log-normal are continuous (PDF).
- The CDF of a continuous variable rises smoothly from 0 to 1.

Sources

- SciPy documentation, `scipy.stats.lognorm` and `scipy.stats.poisson`.

10. Key terms

Term	Meaning
Integration	Finding the area under a curve by adding up infinitely many thin strips
$\int_a^b f(x) dx$	The area under f from a to b ; for a PDF, $P(a \leq X \leq b)$
Log-normal distribution	A right-skewed continuous distribution whose logarithm is normal
Differentiation	Finding the slope of a curve at each point; the derivative of the CDF is the PDF